

Chapter 3: CMOS Transistor Theory

Understanding VLSI from Physical & Mathematical Mechanisms

4-Hour Fast-Track Study Guide | Lecture Reference: UIT Chapter 3 & Weste-Harris CMOS VLSI

4-Hour Time-Boxed Roadmap (Lộ trình học tập 4 giờ)

Follow this exact schedule to achieve complete mastery in 4 hours without wasting time on Chapters 1 & 2:

Time Block	Module Name	Key Mechanism & Outcome
Hour 1 (0h - 1h)	The MOS Capacitor & 3 Modes	Accumulation, Depletion, Inversion, Threshold Voltage V_t origin
Hour 2 (1h - 2h)	Derivation of $I - V$ Equations	Transit time, channel charge, Linear/Saturation math, Pinch-off
Hour 3 (2h - 3h)	pMOS Physics & AMI $0.6\mu\text{m}$ Math	Hole mobility penalty ($\frac{\mu_n}{\mu_p} \approx 2$), $W_p = 2W_n$ sizing, slide 15 math
Hour 4 (3h - 4h)	Parasitic Capacitances & Delay	Gate capacitance C_g , Diffusion capacitance C_{db} , delay formula $\Delta t = \frac{C\Delta V}{I}$

Hour 1: The MOS Capacitor & The 3 Operating Modes

Reference: Slides 1 – 5 | Target Time: 60 minutes

1.1 Beyond the Ideal Switch (Vượt qua tư duy công tắc lý tưởng)

In basic digital logic, transistors are treated as ideal switches: Gate = 1 (closed, zero resistance) and Gate = 0 (open, zero leakage). In real silicon, an ON transistor conducts a **finite current** (dòng điện giới hạn), and its terminals possess **capacitance** (điện dung ký sinh).

⚡ VLSI Design Rule of Thumb (Quy tắc thiết kế vi mạch)

The fundamental law of digital switching speed is:

$$\Delta t = \frac{C}{I} \Delta V$$

To make a chip faster, we must **maximize current I** and **minimize capacitance C** . Current and capacitance together dictate the maximum clock frequency of the processor.

1.2 Physical Structure of the MOS Capacitor

A MOSFET is controlled by a sandwich structure:

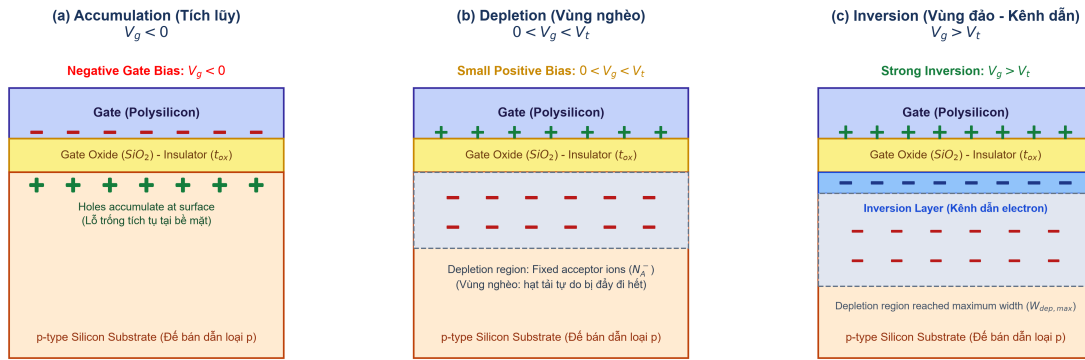
- Gate (Cực Cổng):** Highly conductive Polysilicon (silicon đa tinh thể dẫn điện tốt).
- Dielectric (Lớp điện môi cách điện):** Silicon Dioxide (SiO_2) with thickness t_{ox} and permittivity $\epsilon_{ox} = 3.9\epsilon_0$.
- Body / Substrate (Đế bán dẫn):** p-type silicon doped with Acceptor atoms (N_A , e.g., Boron).

Gate oxide capacitance per unit area is:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14} \text{ F/cm}}{t_{ox}}$$

1.3 The 3 Physical Operating Modes (Ba chế độ hoạt động)

Depending on the gate bias voltage relative to the p-substrate (V_g), the structure exhibits 3 distinct regimes:



Physical Mechanism (Cơ chế vật lý cốt lõi)

1. Accumulation (Chế độ tích lũy - $V_g < 0$ V):

Negative voltage on the gate pulls positive holes (h^+) from the p-substrate directly up to the Si – SiO₂ surface. A dense layer of holes accumulates at the interface. The transistor is strictly OFF.

2. Depletion (Chế độ nghèo - $0 < V_g < V_t$):

A small positive gate voltage repels positive mobile holes away from the surface into the substrate bulk. This exposes immobile negative acceptor ions (N_A^-), creating an insulating depletion zone of thickness W_{dep} containing zero free carriers.

3. Inversion (Chế độ đảo - Hình thành kênh dẫn - $V_g > V_t$):

When V_g exceeds the **Threshold Voltage** (V_t), the electric field becomes strong enough to pull minority electrons (e^-) from the substrate to the surface. The surface silicon inverts from p-type to n-type. This conductive electron layer forms the **Inversion Channel**.

VLSI Design Rule of Thumb (Quy tắc thiết kế vi mạch)

Why Threshold Voltage V_t Exists Physically:

V_t is the exact gate voltage required to achieve **strong inversion**—where the electron concentration at the surface equals the original hole concentration in the substrate ($n_{surface} = N_A$). Below V_t , no conducting channel exists.

Hour 2: Mathematical Derivation of Shockley $I - V$ Equations

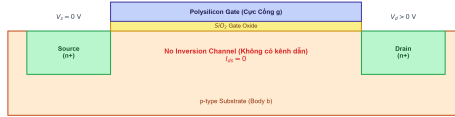
Reference: Slides 6 – 14 | Target Time: 60 minutes

2.1 Terminal Voltages & Conduction Regimes

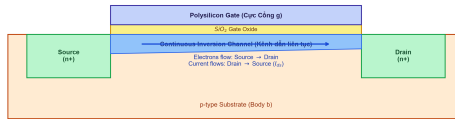
With Source and Body grounded ($V_s = 0$ V, $V_b = 0$ V):

- $V_{gs} = V_g - V_s$: Controls channel inversion.
- $V_{ds} = V_d - V_s$: Creates lateral electric field to pull electrons from Source to Drain.
- $V_{gd} = V_g - V_d = V_{gs} - V_{ds}$: Controls channel inversion depth at the drain end.

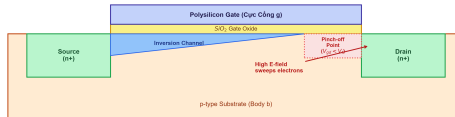
1. Cutoff Region ($V_{gs} < V_t$): Không có kênh dẫn, $I_{ds} = 0$



2. Linear Region ($V_{gs} > V_t$, $V_{ds} < V_{gs} - V_t$): Kênh dẫn liên tục, $I_{ds} \approx V_{ds}$



3. Saturation & Pinch-off ($V_{gs} > V_t$, $V_{ds} \geq V_{gs} - V_t$): Thái kênh tại cực máng, $I_{ds} = I_{ds, sat}$



2.2 First-Principles Derivation of Current (I_{ds})

Current is charge passing through a cross-section per unit time: $I_{ds} = \frac{Q_{\text{channel}}}{t_{\text{transit}}}$.

Mathematical Derivation (Chứng minh toán học)

Step 1: Calculate Total Channel Charge (Q_{channel})

The channel behaves like a parallel-plate capacitor with area $W \times L$. The average channel voltage along the length is

$$V_{\text{channel, avg}} = \frac{V_s + V_d}{2} = \frac{V_{ds}}{2}.$$

The effective gate-to-channel inversion voltage is $V_{\text{eff}} = (V_{gs} - \frac{V_{ds}}{2}) - V_t$.

$$Q_{\text{channel}} = C_{ox} W L \left[(V_{gs} - V_t) - \frac{V_{ds}}{2} \right]$$

Step 2: Calculate Transit Time (t_{transit})

Lateral electric field is $\mathcal{E} = \frac{V_{ds}}{L}$. Electron drift velocity is $v = \mu_n \mathcal{E} = \mu_n \frac{V_{ds}}{L}$.

$$t_{\text{transit}} = \frac{L}{v} = \frac{L^2}{\mu_n V_{ds}}$$

Step 3: Combine to Obtain Linear Current

$$I_{ds} = \frac{Q_{\text{channel}}}{t_{\text{transit}}} = \mu_n C_{ox} \frac{W}{L} \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

2.3 Saturation & Pinch-off Mechanism (Vùng bão hòa & Hiện tượng thắt kênh)

When V_{ds} increases such that $V_{gd} = V_{gs} - V_{ds} \leq V_t$, the inversion charge at the drain edge drops to zero. This is called **Pinch-off** (thắt kênh).

Physical Mechanism (Cơ chế vật lý cốt lõi)

Why Current Does Not Drop to Zero at Pinch-off:

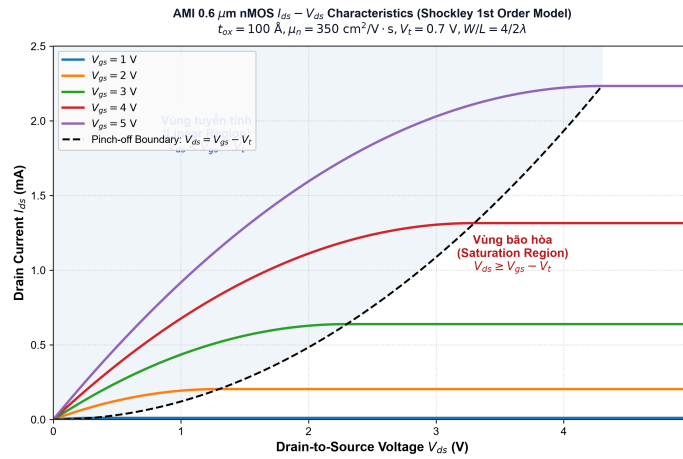
Electrons enter the channel at the Source and travel toward the Drain. At the pinch-off point, electrons are injected into the high-field depletion gap and are **swept (quét mạnh)** into the Drain n^+ diffusion.

Increasing V_{ds} further does not increase channel current because the voltage drop across the active conducting channel remains clamped at $V_{dsat} = V_{gs} - V_t$.

Substituting $V_{ds} = V_{gs} - V_t$ into the linear equation gives the constant saturation current:

$$I_{ds, \text{sat}} = \beta \left[(V_{gs} - V_t)^2 - \frac{(V_{gs} - V_t)^2}{2} \right] = \frac{\beta}{2} (V_{gs} - V_t)^2$$

where $\beta = \mu_n C_{ox} \frac{W}{L}$ is the transistor gain factor.



VLSI Design Rule of Thumb (Quy tắc thiết kế vi mạch)

Shockley 1st-Order Summary:

- **Cutoff** ($V_{gs} < V_t$): $I_{ds} = 0$
- **Linear** ($V_{gs} \geq V_t$, $V_{ds} < V_{gs} - V_t$): $I_{ds} = \beta \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$
- **Saturation** ($V_{gs} \geq V_t$, $V_{ds} \geq V_{gs} - V_t$): $I_{ds} = \frac{\beta}{2} (V_{gs} - V_t)^2$

Hour 3: pMOS Physics & AMI 0.6 μm Hand Calculation

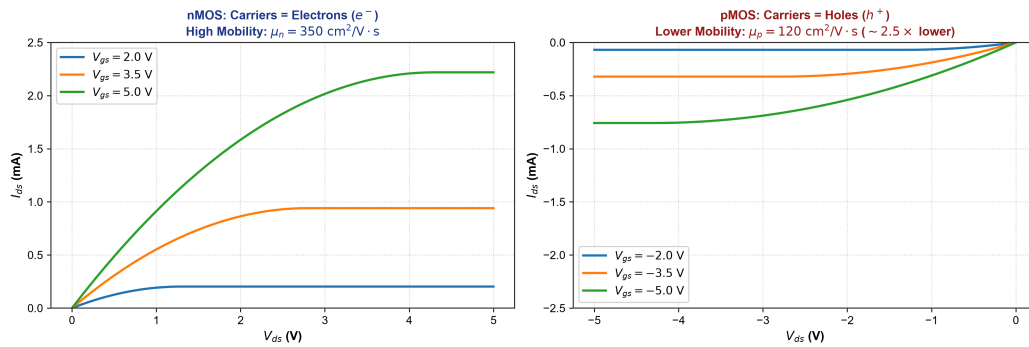
Reference: Slides 15 – 17 | Target Time: 60 minutes

3.1 pMOS Physics: Polarity Inversion & Mobility Penalty

A pMOS transistor is formed inside an n-well on a p-substrate, using p^+ source/drain diffusions. Current conduction is carried by **holes** (h^+) instead of electrons:

- All voltages and currents have **negative polarity**: $V_{gs} < 0$, $V_{ds} < 0$, $I_{ds} < 0$.
- Cutoff occurs when $V_{gs} > V_{tp}$ (where $V_{tp} \approx -0.7\text{ V}$).

nMOS vs pMOS Comparison: Polarity Inversion and Hole Mobility Penalty
(Để có cùng dòng điện, pMOS phải có bề rộng $W_p \approx 2 - 3 \times W_n$)



Physical Mechanism (Cơ chế vật lý cốt lõi)

Why Are Holes 2 to 3 Times Slower Than Electrons?

Electrons move through the conduction band (dải dẫn) with low effective mass and few collisions. In contrast, holes move through the valence band (dải hóa trị) as electrons jump from one covalent bond to another. This higher scattering results in:

$$\frac{\mu_n}{\mu_p} \approx 2 \text{ to } 3 \quad \left(\mu_n \approx 350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}, \mu_p \approx 120 \frac{\text{cm}^2}{\text{V} \cdot \text{s}} \right)$$

VLSI Design Rule of Thumb (Quy tắc thiết kế vi mạch)

The Inverter Sizing Golden Rule ($\frac{W_p}{W_n} = 2$):

Because holes travel half as fast as electrons, a pMOS transistor must be designed with **twice the channel width** ($W_p \approx 2W_n$) of an nMOS transistor to deliver the same drive current, ensuring symmetric rise and fall propagation delays ($t_{pLH} \approx t_{pHL}$).

3.2 Worked Hand Calculation: AMI 0.6 μ m Process (Directly solving Slide 15 from the university lecture)

Given Parameters:

- Process Node: 0.6 μ m, Gate oxide: $t_{ox} = 100 \text{ \AA} = 100 \times 10^{-8} \text{ cm} = 10 \text{ nm}$.
- Electron mobility: $\mu_n = 350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$.
- Threshold voltage: $V_t = 0.7 \text{ V}$. Transistor geometry: $\frac{W}{L} = \frac{4\lambda}{2\lambda} = 2$.

Mathematical Derivation (Chứng minh toán học)

Step 1: Calculate Oxide Capacitance per Area (C_{ox})

$$\varepsilon_{ox} = 3.9 \times \varepsilon_0 = 3.9 \times (8.85 \times 10^{-14} \text{ F/cm}) = 3.45 \times 10^{-13} \text{ F/cm}$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-13} \text{ F/cm}}{100 \times 10^{-8} \text{ cm}} = 3.45 \times 10^{-7} \text{ F/cm}^2$$

Step 2: Calculate Process Transconductance Parameter ($k'_n = \mu_n C_{ox}$)

$$k'_n = \mu_n C_{ox} = 350 \frac{\text{cm}^2}{\text{V} \cdot \text{s}} \times (3.45 \times 10^{-7} \text{ F/cm}^2) = 120.8 \mu\text{A/V}^2$$

Step 3: Calculate Transistor Gain Factor (β)

$$\beta = k'_n \frac{W}{L} = 120.8 \mu\text{A/V}^2 \times 2 = 241.6 \mu\text{A/V}^2 = 0.242 \text{ mA/V}^2$$

Step 4: Calculate Saturation Current at $V_{gs} = 5 \text{ V}$

$$V_{dsat} = V_{gs} - V_t = 5.0 \text{ V} - 0.7 \text{ V} = 4.3 \text{ V}$$

$$I_{ds, \text{sat}} = \frac{\beta}{2} (V_{gs} - V_t)^2 = \frac{0.242 \text{ mA/V}^2}{2} \times (4.3 \text{ V})^2 = 0.121 \times 18.49 = 2.24 \text{ mA}$$

Comparison: pMOS Equivalent for AMI 0.6 μ m

For a pMOS in the same process ($\mu_p = 120 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$, $V_{tp} = -0.7 \text{ V}$, $\frac{W}{L} = 2$):

$$\beta_p = \mu_p C_{ox} \frac{W}{L} = 120 \times (3.45 \times 10^{-7}) \times 2 \approx 0.083 \text{ mA/V}^2$$

At $V_{gs} = -5 \text{ V}$:

$$I_{ds, \text{sat}} = -\frac{\beta_p}{2} (|V_{gs}| - |V_{tp}|)^2 = -\frac{0.083}{2} \times (4.3)^2 \approx -0.77 \text{ mA}$$

Notice that $\frac{I_{ds, n}}{|I_{ds, p}|} = \frac{2.24}{0.77} \approx 2.9 \times$, mathematically proving why pMOS must be wider!

Hour 4: Transistor Parasitic Capacitance & Circuit Speed

Reference: Slides 18 – 20 | Target Time: 60 minutes

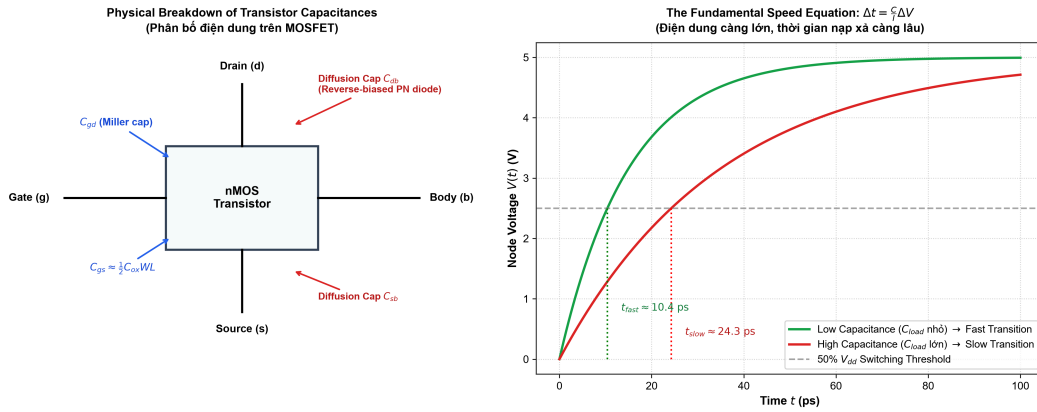
4.1 Gate Capacitance (C_g)

Gate capacitance creates the inversion channel and acts as the capacitive load on the preceding driver:

$$C_g = C_{ox}WL = C_{\text{per_micron}}W$$

In submicron processes, gate capacitance is approximately:

$$C_{\text{per_micron}} = C_{ox}L_{\text{min}} \approx 1.5 \text{ to } 2.0 \text{ fF}/\mu\text{m}$$



4.2 Parasitic Diffusion Capacitance (C_{sb}, C_{db})

The n^+ source and drain regions form **reverse-biased PN junction diodes** with the p-substrate:

- Diffusion capacitance is **purely parasitic** (wastes dynamic power and increases delay).
- Composed of **Bottom plate capacitance** ($C_{\text{bottom}} \propto W \times L_d$) and **Sidewall capacitance** ($C_{\text{sidewall}} \propto 2W + 2L_d$).
- Contacted diffusion capacitance is roughly equal to C_g . Uncontacted shared diffusion drops to $\approx 0.5C_g$.

4.3 Fundamental Delay Equation

When a gate switches, current charges the load capacitance $C_L = C_{\text{gate}} + C_{\text{diff}} + C_{\text{wire}}$:

$$\Delta t = \frac{C_L \Delta V}{I_{\text{avg}}}$$

To double chip clock frequency, we must either double saturation current I or cut load capacitance C_L in half!

4.4 Summary Cheat Sheet (Bảng tóm tắt toàn chương)

Symbol / Parameter	Physical Meaning	Typical Value / Formula
C_{ox}	Gate oxide capacitance per area	$\frac{\epsilon_{ox}}{t_{ox}} \approx 3.45 \times 10^{-7} \text{ F/cm}^2$
β	Transconductance gain factor	$\mu C_{ox} \left(\frac{W}{L} \right)$
Linear Region	Continuous resistive channel	$V_{gs} \geq V_t$ and $V_{ds} < V_{gs} - V_t$
Saturation Region	Channel pinched-off at drain	$V_{gs} \geq V_t$ and $V_{ds} \geq V_{gs} - V_t$
$I_{ds, \text{sat}}$	Shockley saturation current	$\frac{\beta}{2} (V_{gs} - V_t)^2$
$\frac{\mu_n}{\mu_p}$	Mobility ratio (electrons vs holes)	$\approx 2 \text{ to } 3 \text{ } (\rightarrow W_p = 2W_n)$
Δt	Gate propagation delay	$\frac{C_L \Delta V}{I_{\text{avg}}}$

4.5 10-Minute Self-Assessment Quiz (Bài kiểm tra nhanh)

- Q1:** An nMOS transistor has $V_{gs} = 3.0 \text{ V}$, $V_t = 0.6 \text{ V}$, and $V_{ds} = 1.5 \text{ V}$. Which operating region is it in?
Answer: $V_{\text{dsat}} = V_{gs} - V_t = 3.0 - 0.6 = 2.4 \text{ V}$. Since $V_{ds} = 1.5 \text{ V} < 2.4 \text{ V}$, the channel is continuous and the transistor is in the **Linear Region**.
- Q2:** Why does drain current remain constant in the saturation region even if V_{ds} increases?
Answer: The channel pinches off at the drain side. The voltage across the active inversion channel is clamped at $V_{\text{dsat}} = V_{gs} - V_t$. Extra voltage drops across the pinch-off depletion gap without increasing the channel field.
- Q3:** Why is pMOS designed twice as wide as nMOS ($W_p \approx 2W_n$) in standard CMOS cells?
Answer: Hole mobility in silicon is $2 - 3 \times$ lower than electron mobility ($\frac{\mu_n}{\mu_p} \approx 2 - 3$). A $2 \times$ wider pMOS matches nMOS drive current, achieving symmetric rise and fall delays ($t_{pLH} \approx t_{pHL}$).
- Q4:** Why does sharing diffusion nodes in layout improve circuit speed?
Answer: Shared diffusion eliminates contact spacing, reducing junction area by $\approx 50\%$. This reduces parasitic diffusion capacitance C_{db} , decreasing propagation delay $\Delta t = \frac{C \Delta V}{I}$.